

PPHS 101: Introduction to Population Science

Module 3: Population Structure

Md. Sanaul Haque Mondal
Department of Social Relations
East West University, Bangladesh

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Concepts and definitions

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Characteristics or attributes of population are classified into two groups:

A) Ascribed characteristics:

- biologically or culturally assigned and the individual cannot exercise his/her choice.
- Examples: age, sex, race, and mother tongue.
- nationality, religion, caste
(be later changed)

B) Achieved characteristics

- open to the individual's choice.
- Examples: marital status, educational attainment, labor forced status, occupation

Source: Bhende & Kanitkar (1978)

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Subgroup

What is the proportional distribution of males and females in a given population?

How are children, young adults, and elderly persons distributed in a population?

What percentage of a population is illiterate?

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*Subgroup (continue)***Examples**

- Sex :
 - entire population may be classified into two groups:
 - males and females
- Religion
 - entire population may be divided into different religion groups:
 - Muslim, Hindu, Christian, etc.

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Importance of age and sex

- Division of labor: based on age and sex.
- Age distribution have consequences for educational, political, and economic life
- Socioeconomic and demographic development
- Labor-force participation and gender relations
- There is a close relationship between *demographic variables* and *demographic processes*.

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Sex and Gender

- Sex:
 - biological characteristics of males and females (ascribed characteristics)
- Gender:
 - psychological and social characteristics arising from belief systems of what male and female behaviour is or should be (achieved characteristic)

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- **Determination of sex:**
 - biological and social considerations
- **Census/survey:**
 - self-identification of his or her sex, not on biological considerations

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Biological definitions of sex:

- Sex is based on chromosomes (XX/ XY)
- Sex is based on gonads (testes in males and ovaries in females)
- Sex is based on sex-specific hormones (Androgens, Testosterone)
- Sex is based on internal sexual properties (two sets of ducts)
- Sex-specific external genitals (assignment of sex at the birth of the baby)

Source: Poston and Bouvier, 2010

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Age

- Ascribed, yet changeable, characteristic
- Number of years after birth
- Definition: the estimated or calculated interval of time between the date of birth and the date of the census, expressed in complete solar years (United Nations, 1998).

Source: Bhende & Kanitkar (1978)

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Infant	a person less than 1 year of age
Neonatal	first 28 days after birth
Child	less than 15 years of age (child mortality refers to deaths of children under 5 years of age)
Old	those who are 65 years of age and over.
Adult	those who are 15 years of age and over
Adolescent	age range of about 13-19 years of age

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Methods for analyzing age and sex distributions

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A. The population pyramid

- A graphic representation of the age/sex structure of the population
- Also known as age/sex pyramid
- Shows for a specific point in time the different surviving cohorts of persons of each sex

Source: Poston and Bouvier, 2010

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• Characteristics:

- Two ordinary histograms (bar graphs), representing the male and female populations in, usually, 1- or 5-year age categories, placed on their sides and back to back.
- The base of the pyramid, representing the size of each of the age/sex population groups, is presented in either absolute numbers or in percentages.

Source: Poston and Bouvier, 2010

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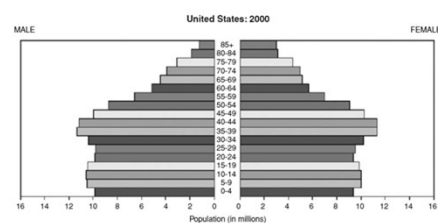


Figure: Age/sex pyramid, United States, 2000

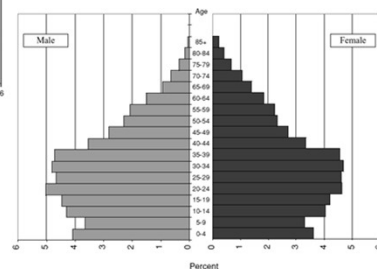


Figure: Population pyramid, Republic of Korea, 1995

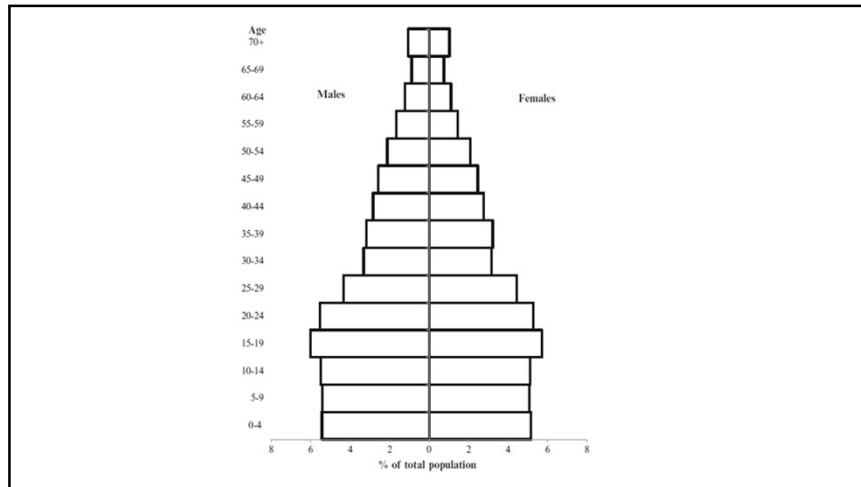
Source: Poston and Bouvier, 2010

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Exercise

Age group	Population		% of total population	
	Males	Females	Males	Females
(1)	(2)	(3)	(4) = (2)*100/78,728,329	(5) = (3)*100/78,728,329
0-4	4,282,316	4,071,818	5.44	5.17
5-9	4,265,412	4,007,172	5.42	5.09
10-14	4,330,443	4,023,320	5.50	5.11
15-19	4,738,218	4,500,674	6.02	5.72
20-24	4,358,058	4,155,045	5.54	5.28
25-29	3,411,636	3,498,432	4.33	4.44
30-34	2,613,911	2,502,494	3.32	3.18
35-39	2,498,485	2,535,871	3.17	3.22
40-44	2,236,659	2,186,333	2.84	2.78
45-49	2,029,328	1,941,694	2.58	2.47
50-54	1,667,511	1,639,886	2.12	2.08
55-59	1,312,016	1,135,920	1.67	1.44
60-64	971,087	871,006	1.23	1.11
65-69	692,772	597,495	0.88	0.76
70+	842,588	810,729	1.07	1.03
Total ^a	40,250,440	38,477,889	51.13	48.87

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Exercise 2: Identify the characteristics:

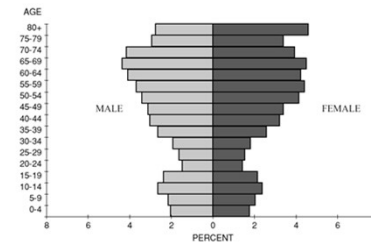


Figure. Population by age and sex: Llanos County, Texas, 2000

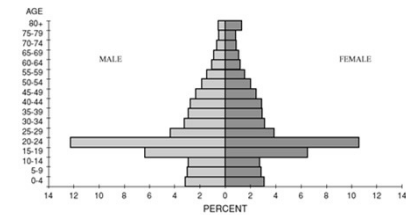


Figure. Population by age and sex: Brazos County, Texas, 2000

Source: Poston and Bouvier, 2010

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B. Age dependency

Dependency ratio (DR):

- The DR is the ratio of the dependent-age population (both young [persons 0-14 years old] and old [persons 65 years of age and older]) to the working-age population (persons 15-64 years old).
- DR is multiplied by 100
- The higher the ratio, the more people each worker has to support; the lower the DR, the fewer the number of dependents.

Source: Poston and Bouvier, 2010

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$$\text{Total (Age) Dependency Ratio} = \frac{(\text{Population 0-14} + \text{Population 64+})}{\text{Working age population 15-64}} \times 100$$

$$\text{Child Dependency Ratio} = \frac{\text{Population 0-14}}{\text{Working age population 15-64}} \times 100$$

$$\text{Old-Age Dependency Ratio} = \frac{\text{Population 64+}}{\text{Working age population 15-64}} \times 100$$

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Exercise 4:

- Population ages 0-15 years= 230,416
- Population ages 65-plus years= 15,060
- Population ages 16-64 years= 301,703

Calculate:

- Total population
- Youth dependency ratio
- Old-age dependency ratio
- Total dependency ratio

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$$\text{Dependency Ratio} = \frac{230,416 + 15,060}{301,703} \times 100$$

$$= 81.36$$

$$\text{Young Dependency Ratio} = \frac{230,416}{301,703} \times 100$$

$$= 76.37$$

$$\text{Old Dependency Ratio} = \frac{15,060}{301,703} \times 100$$

$$= 4.99$$

$$\text{Total Dependency Ratio} = \text{Young Dependency Ratio} + \text{Old Dependency Ratio}$$

$$\text{Hence } 81.36 = 76.37 + 4.99$$

Source: Bhende & Kanitkar (1978)

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Table. Values of youth-dependency ratio, old-age dependency ratio, and total dependency ratio, selected countries of the world, 2001

Country	Youth-DR	Old-Age-DR	Total DR
South Korea	31.0	9.9	40.9
China	32.9	10.0	42.9
Italy	20.6	26.5	47.1
Spain	22.1	25.0	47.1
Japan	22.1	25.0	47.1
United States	31.8	19.7	51.5
Yugoslavia	31.8	19.7	51.5
Sweden	29.7	26.6	56.3
Mexico	55.7	8.2	63.9
Nigeria	83.0	5.7	88.7
Yemen	98.0	6.1	104.1
Niger	104.2	4.2	108.4
Uganda	108.5	4.3	112.8

Source: Population Reference Bureau, 2001.

Source: Bhende & Kanitkar (1978)

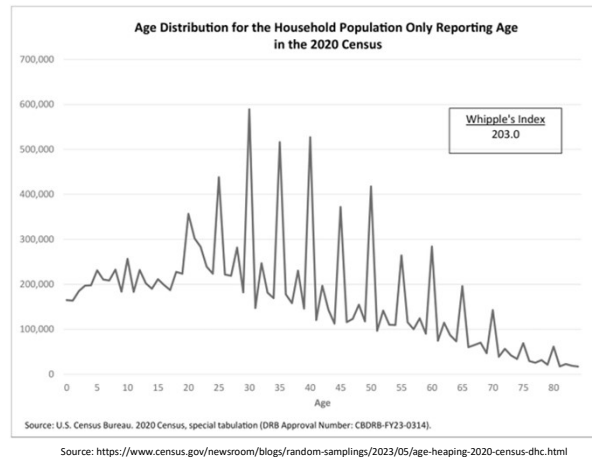
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C. Age heaping

- If a population tends to report certain ages (say, those ending in 0 or 5) at the expense of other ages, this is known as age heaping.
- Age heaping tends to be more pronounced among populations or population subgroups with low levels of education
- The causes and patterns of [age heaping] vary from culture to culture, but preference for ages ending in '0' and in '5' is quite widespread

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- Age heaping is easily detected using graphs and indices e.g., Whipple's method (WM)



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D. Sex structure

Methods to index sex composition

- 1) the masculinity proportion,
- 2) the ratio of the excess or deficit of males to the total population, and
- 3) the sex ratio.

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Factors affecting sex ratio

- Sex ratio of newborn babies or the sex ratio at birth
- Sex ratio of deceased persons
- Sex ratio of net migrants

Source: Bhende & Kanitkar (1978)

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- Homework/Group discussion:

Factors affecting sex ratio in Bangladesh

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